

Warm-up (5 min)

1. Quick recall: state the four kinematics equations from last session. $v = u + at$ · $s = ut + \frac{1}{2}at^2$ · $v^2 = u^2 + 2as$ · $a = (v - u)/t$
2. Hook question: “If you fire a bullet horizontally over level ground, and at the exact same instant drop a second bullet from the same height, which one hits the ground first?” Answer to draw out: **they land at the same time**. The horizontal motion of the fired bullet doesn’t affect its vertical fall — gravity acts on both identically. This is the whole idea behind today’s topic.

Concept Notes (15 min)

Nothing below is a formula to memorize cold — every one is derived from the same four SUVAT equations from Session 1. Walk through each derivation with him; that’s the actual point of this section.

The core idea

A projectile’s motion is two independent 1D motions happening at the same time. Why independent? Gravity only pulls straight down — there’s no horizontal force acting on the projectile (ignoring air resistance).

Newton’s Second Law of Motion: The acceleration produced by a net force on an object is directly proportional to the magnitude of the net force, inversely proportional to the object’s mass, and in the same direction as the net force. Mathematically: $F_{\text{net}} = ma$, equivalently $a = F_{\text{net}} / m$, where F_{net} is the net (total) force in newtons (N), m is mass in kilograms (kg), and a is acceleration in m/s^2 . Applied here: horizontally, $F_{\text{net},x} = 0$ (no horizontal force), so $a_x = 0/m = 0$ — zero horizontal acceleration, which is exactly why the horizontal velocity stays constant throughout the flight.

- **Horizontal (x):** zero acceleration → **constant velocity**.
- **Vertical (y):** constant acceleration $g = 9.8 \text{ m/s}^2$, downward — this is exactly the free-fall motion from Session 1, just happening at the same time as the horizontal motion instead of on its own.

They don’t affect each other, which is why we can reuse the Session 1 SUVAT equations on each direction separately.

Case A — Horizontal Launch (rolls/dropped off a ledge, table, cliff) — full derivation

Setup: launched purely horizontally with speed v_0 , so the initial vertical velocity is 0. Take downward as positive for the y-direction here, and the launch direction as positive for x.

Deriving x (horizontal distance): Start from the Session 1 equation $s = ut + \frac{1}{2}at^2$, applied to the x-direction. Horizontal acceleration $a = 0$ (no horizontal force), and initial horizontal speed $u = v_0$: $x = v_0t + \frac{1}{2}(0)t^2 \rightarrow x = v_0t$

Deriving y (vertical distance fallen): Same equation, applied to the y-direction. Initial vertical speed $u = 0$ (purely horizontal launch), and vertical acceleration $a = g$: $y = (0)t + \frac{1}{2}gt^2 \rightarrow y = \frac{1}{2}gt^2$

Deriving v_y (vertical velocity at time t): Start from $v = u + at$, applied to the y-direction, with $u = 0$, $a = g$: $v_y = 0 + gt \rightarrow v_y = gt$

Deriving the time to land from height h: “Landing” means the object has fallen the full height, so $y = h$. Substitute into the y equation we just derived and solve for t: $h = \frac{1}{2}gt^2 \rightarrow t^2 = 2h/g \rightarrow t = \sqrt{2h/g}$

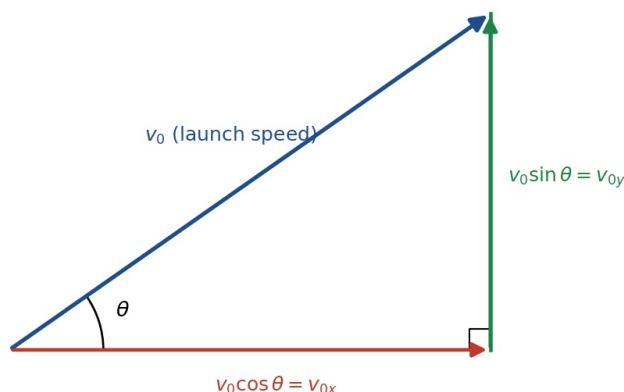
So every formula in Case A is just the two Session-1 equations ($s = ut + \frac{1}{2}at^2$ and $v = u + at$) applied once horizontally with $a = 0$, and once vertically with $u = 0$, $a = g$.

Case B — Launched at an Angle (kicked/thrown upward at angle θ , lands at the same height it launched from) — full derivation

Deriving v_{0x} and v_{0y} (splitting the launch velocity into components) — full SOH-CAH-TOA derivation:

The launch velocity v_0 points at angle θ above horizontal — it's a single vector, not already separated into "horizontal" and "vertical" pieces. To split it, draw the right triangle it forms with the horizontal (this is exactly the vector-component method from Hewitt Ch. 6.6):

Resolving the launch velocity v_0 into components (SOH-CAH-TOA)



Right triangle showing v_0 resolved into v_{0x} and v_{0y}

- **Step 1 — construct the triangle.** Draw v_0 as an arrow from the origin at angle θ above the horizontal. Drop a vertical line from the tip of that arrow down to the horizontal axis. This creates a right triangle where:
 - the **hypotenuse** is v_0 itself (length = the launch speed)
 - the **horizontal leg** (along the ground) is v_{0x}
 - the **vertical leg** (the dropped line) is v_{0y}
 - the right angle sits where the vertical leg meets the horizontal leg
- **Step 2 — identify opposite, adjacent, hypotenuse relative to θ .** θ is the angle at the origin, between the horizontal leg and the hypotenuse.
 - The side **opposite** θ (doesn't touch the angle) is the vertical leg $\rightarrow v_{0y}$
 - The side **adjacent** to θ (touches the angle, isn't the hypotenuse) is the horizontal leg $\rightarrow v_{0x}$
 - The **hypotenuse** is v_0
- **Step 3 — recall SOH-CAH-TOA.** SOH: $\sin\theta = \text{Opposite} / \text{Hypotenuse}$ CAH: $\cos\theta = \text{Adjacent} / \text{Hypotenuse}$ TOA: $\tan\theta = \text{Opposite} / \text{Adjacent}$
- **Step 4 — apply SOH to solve for v_{0y} .** $\sin\theta = v_{0y} / v_0$ Multiply both sides by v_0 : $v_{0y} = v_0 \sin\theta$
- **Step 5 — apply CAH to solve for v_{0x} .** $\cos\theta = v_{0x} / v_0$ Multiply both sides by v_0 : $v_{0x} = v_0 \cos\theta$
- **Step 6 — sanity-check with the Pythagorean theorem.** The two components should recombine into the original speed: $v_{0x}^2 + v_{0y}^2 = v_0^2$. Substitute: $(v_0 \cos\theta)^2 + (v_0 \sin\theta)^2 = v_0^2 \cos^2\theta + v_0^2 \sin^2\theta = v_0^2 (\cos^2\theta + \sin^2\theta)$ Using the identity $\sin^2\theta + \cos^2\theta = 1$, this equals $v_0^2 (1) = v_0^2$ ✓ — confirms the components are correct.
- **Worked numeric check:** $v_0 = 20 \text{ m/s}$, $\theta = 30^\circ \rightarrow v_{0x} = 20 \cos 30^\circ = 17.32 \text{ m/s}$, $v_{0y} = 20 \sin 30^\circ = 10 \text{ m/s}$. Check: $17.32^2 + 10^2 = 300 + 100 = 400 = 20^2$ ✓.

v_{0x} stays constant for the whole flight (horizontal acceleration = 0, from Newton's Second Law above);

v_{0y} is just the *starting* vertical speed — gravity changes it as time passes, exactly like Case A's $v_y = gt$, except here it starts at v_{0y} instead of 0.

Deriving $x(t)$, $y(t)$, $v_y(t)$: Same approach as Case A, but now the vertical direction starts with a nonzero speed and we take upward as positive, so $a = -g$: - Horizontal ($u = v_{0x}$, $a = 0$): $x(t) = v_{0x} \cdot t$ - Vertical ($u = v_{0y}$, $a = -g$), from $s = ut + \frac{1}{2}at^2$: $y(t) = v_{0y} \cdot t - \frac{1}{2}gt^2$ - Vertical velocity, from $v = u + at$: $v_y(t) = v_{0y} - gt$

Deriving the time to reach the peak: At the very top of the arc, the projectile is momentarily neither rising nor falling, so $v_y = 0$. Set the $v_y(t)$ equation to zero and solve for t : $0 = v_{0y} - g \cdot t_{\text{up}} \rightarrow t_{\text{up}} = v_{0y}/g$

Deriving the max height: Use the third Session-1 equation, $v^2 = u^2 + 2as$, applied vertically between launch and the peak, where the vertical velocity is 0: $0 = v_{0y}^2 - 2gH \rightarrow H = v_{0y}^2 / (2g)$

Deriving the total time of flight: “Lands at launch height” means y returns to 0. Set the $y(t)$ equation to zero and solve for t : $0 = v_{0y} \cdot T - \frac{1}{2}gT^2 \rightarrow 0 = T(v_{0y} - \frac{1}{2}gT)$ This is zero when $T = 0$ (the launch instant itself) or when $v_{0y} - \frac{1}{2}gT = 0$, i.e. $T = 2v_{0y}/g$. The second solution is the landing time we want: $T = 2v_{0y}/g$ (Notice this is exactly $2 \times t_{\text{up}}$ — confirming what Session 1's symmetric free-fall behavior already told us: time going up equals time coming down.)

Deriving the range: Range is just the horizontal distance covered in the total flight time: $R = x(T) = v_{0x} \cdot T$. Substitute $T = 2v_{0y}/g$: $R = v_{0x} \cdot (2v_{0y}/g) = 2v_{0x}v_{0y}/g$ Now substitute the components from the first derivation, $v_{0x} = v_0 \cos \theta$ and $v_{0y} = v_0 \sin \theta$: $R = 2v_0^2 \sin \theta \cos \theta / g$ Using the double-angle identity $2 \sin \theta \cos \theta = \sin(2\theta)$ (worth a quick note if he hasn't seen it yet): $R = v_0^2 \sin(2\theta) / g$

So every Case B formula also traces straight back to the three Session-1 SUVAT equations — the only new ingredient is splitting v_0 into components with right-triangle trig at the start.

The strategy for any projectile problem

1. Split the initial velocity into horizontal and vertical components (skip this step for a horizontal launch — it's already just v_0 horizontal, 0 vertical).
2. Solve the **vertical** motion first — it's the one gravity constrains, so it usually gives you time.
3. Plug that time into the **horizontal** equation to get distance, or vice versa.
4. If asked for a resultant velocity (speed and direction) at some instant, find v_x and v_y separately, then combine: $\text{speed} = \sqrt{v_x^2 + v_y^2}$.

Worked Examples (10 min)

Example 1 (horizontal launch): A ball rolls off a table 1.25 m high with a horizontal speed of 3 m/s. Find (a) time to hit the floor, (b) horizontal distance traveled, (c) vertical velocity at impact.

- a. $t = \sqrt{2 \times 1.25 / 9.8} = \sqrt{0.255} = 0.51 \text{ s}$
- b. $x = 3 \times 0.51 = 1.52 \text{ m}$
- c. $v_y = 9.8 \times 0.51 = 4.95 \text{ m/s}$

Example 2 (angled launch): A stone is thrown at 20 m/s at 30° above horizontal from ground level. Find (a) time of flight, (b) max height, (c) range.

- $v_{0x} = 20 \cos 30^\circ = 17.32 \text{ m/s}$, $v_{0y} = 20 \sin 30^\circ = 10 \text{ m/s}$
- a. $T = 2 \times 10 / 9.8 = 2.04 \text{ s}$
- b. $H = 10^2 / (2 \times 9.8) = 5.10 \text{ m}$
- c. $R = 17.32 \times 2.04 = 35.3 \text{ m}$ (check: $400 \times \sin 60^\circ / 9.8 = 35.3 \text{ m}$ ✓)

Practice Problems

10 numerical questions, increasing difficulty. Use $g = 9.8 \text{ m/s}^2$. Do Q1–Q4 together in class, assign the rest as homework.

- Horizontal launch — time:** A marble rolls off a 0.8 m high desk at a horizontal velocity of 2 m/s. Find the time to hit the floor.
- Horizontal launch — range:** Using Q1, find the horizontal distance the marble travels before landing.
- Horizontal launch — impact speed:** A ball leaves a table horizontally at 5 m/s from a height of 2.0 m. Find the vertical velocity just before impact, and the resultant (total) speed at impact.
- Angled launch — time of flight:** A ball is kicked at 15 m/s at 40° above horizontal on level ground. Find the time of flight.
- Angled launch — max height:** For the same kick as Q4, find the maximum height reached.
- Angled launch — range:** For the same kick as Q4, find the horizontal range.
- Reverse problem:** A projectile launched at 45° lands 20 m away. Find its initial launch speed. (Hint: at 45° , $\sin(2\theta) = 1$.)
- Cliff problem:** A stone is thrown horizontally from the top of a 45 m cliff at 8 m/s. Find (a) time to hit the ground, (b) horizontal distance traveled, (c) speed just before impact.
- Velocity at a given instant:** A ball is thrown at 25 m/s at 53° above horizontal. Find its horizontal and vertical velocity components 2 seconds after launch, and state whether it has passed its highest point yet.
- Challenge — complementary angles:** A projectile is launched at 18 m/s, once at 30° and once at 60° . Calculate the range for each and confirm they're equal. Then calculate the time of flight for each and explain why they're different even though the range is the same.

Answer Key — Full Step-by-Step Solutions

1. Horizontal launch — time Given: $h = 0.8 \text{ m}$, $g = 9.8 \text{ m/s}^2$ (the horizontal speed 2 m/s doesn't matter for this part — only vertical motion determines fall time). - Step 1: Vertical motion starts from rest ($v_{0y} = 0$), so use $y = \frac{1}{2}gt^2$. - Step 2: Solve for t : $t = \sqrt{2y/g}$. - Step 3: Substitute: $t = \sqrt{(2 \times 0.8) / 9.8} = \sqrt{1.6/9.8} = \sqrt{0.1633}$. - Step 4: $t = 0.40 \text{ s}$. **Answer: $t \approx 0.40 \text{ s}$**

2. Horizontal launch — range Given: $v_0 = 2 \text{ m/s}$ (horizontal), $t = 0.40 \text{ s}$ (from Q1). - Step 1: Horizontal velocity is constant (no horizontal acceleration), so use $x = v_0 \cdot t$. - Step 2: Substitute: $x = 2 \times 0.40$. - Step 3: $x = 0.81 \text{ m}$ (using the more precise $t = 0.404 \text{ s}$). **Answer: $x \approx 0.81 \text{ m}$**

3. Horizontal launch — impact speed Given: $h = 2.0 \text{ m}$, $v_0 = 5 \text{ m/s}$ (horizontal), $g = 9.8 \text{ m/s}^2$. - Step 1: Find time to fall: $t = \sqrt{2h/g} = \sqrt{(4/9.8)} = \sqrt{0.408} = 0.64 \text{ s}$. - Step 2: Find vertical velocity at impact: $v_y = gt = 9.8 \times 0.64 = 6.26 \text{ m/s}$. - Step 3: Horizontal velocity is unchanged: $v_x = 5 \text{ m/s}$. - Step 4: Combine components into the resultant speed: $\text{speed} = \sqrt{v_x^2 + v_y^2} = \sqrt{(5^2 + 6.26^2)} = \sqrt{(25 + 39.2)} = \sqrt{64.2}$. - Step 5: $\text{speed} = 8.01 \text{ m/s}$. **Answer: $t = 0.64 \text{ s}$, $v_y = 6.26 \text{ m/s}$, resultant speed $\approx 8.01 \text{ m/s}$**

4. Angled launch — time of flight Given: $v_0 = 15 \text{ m/s}$, $\theta = 40^\circ$, level ground. - Step 1: Find the initial vertical component: $v_{0y} = v_0 \sin \theta = 15 \times \sin 40^\circ = 15 \times 0.643 = 9.64 \text{ m/s}$. - Step 2: Time of flight (lands at launch height): $T = 2v_{0y}/g$. - Step 3: Substitute: $T = 2 \times 9.64 / 9.8 = 19.28/9.8$. - Step 4: $T = 1.97 \text{ s}$. **Answer: $T \approx 1.97 \text{ s}$**

5. Angled launch — max height Given: $v_{0y} = 9.64 \text{ m/s}$ (from Q4), $g = 9.8 \text{ m/s}^2$. - Step 1: Use $H = v_{0y}^2 / (2g)$. - Step 2: Substitute: $H = 9.64^2 / (2 \times 9.8) = 92.93/19.6$. - Step 3: $H = 4.74 \text{ m}$. **Answer: $H \approx 4.74 \text{ m}$**

6. Angled launch — range Given: $v_0 = 15 \text{ m/s}$, $\theta = 40^\circ$, $T = 1.97 \text{ s}$ (from Q4). - Step 1: Find the horizontal component: $v_{0x} = v_0 \cos \theta = 15 \times \cos 40^\circ = 15 \times 0.766 = 11.49 \text{ m/s}$. - Step 2: Range = horizontal speed $\times$ total time: $R = v_{0x} \times T = 11.49 \times 1.97$. - Step 3: $R = 22.6 \text{ m}$. - Step 4 (check with the shortcut formula): $R = v_0^2 \sin(2\theta) / g = 225 \times \sin 80^\circ / 9.8 = 225 \times 0.985/9.8 = 22.6 \text{ m}$ ✓ matches. **Answer: $R \approx 22.6 \text{ m}$**

7. Reverse problem — find launch speed from range Given: $R = 20 \text{ m}$, $\theta = 45^\circ$. - Step 1: Start from $R = v_0^2 \sin(2\theta) / g$. - Step 2: At 45° , $2\theta = 90^\circ$ and $\sin 90^\circ = 1$, so the formula simplifies to $R = v_0^2 / g$. - Step 3: Solve for v_0 : $v_0 = \sqrt{Rg}$. - Step 4: Substitute: $v_0 = \sqrt{20 \times 9.8} = \sqrt{196}$. - Step 5: $v_0 = 14 \text{ m/s}$. **Answer: $v_0 = 14 \text{ m/s}$**

8. Cliff problem Given: $h = 45 \text{ m}$, $v_0 = 8 \text{ m/s}$ (horizontal), $g = 9.8 \text{ m/s}^2$. - Step 1 (time): $t = \sqrt{2h/g} = \sqrt{(90/9.8)} = \sqrt{9.184}$. - Step 2: $t = 3.03 \text{ s}$. - Step 3 (horizontal distance): $x = v_0 \times t = 8 \times 3.03$. - Step 4: $x = 24.2 \text{ m}$. - Step 5 (vertical velocity at impact): $v_y = gt = 9.8 \times 3.03 = 29.7 \text{ m/s}$. - Step 6 (resultant impact speed): $\text{speed} = \sqrt{v_x^2 + v_y^2} = \sqrt{8^2 + 29.7^2} = \sqrt{64 + 882.1} = \sqrt{946.1}$. - Step 7: $\text{speed} = 30.8 \text{ m/s}$. **Answer: $t \approx 3.03 \text{ s}$, $x \approx 24.2 \text{ m}$, impact speed $\approx 30.8 \text{ m/s}$**

9. Velocity at a given instant Given: $v_0 = 25 \text{ m/s}$, $\theta = 53^\circ$, $t = 2 \text{ s}$. - Step 1: Horizontal component (constant for the whole flight): $v_x = v_0 \cos \theta = 25 \times \cos 53^\circ = 25 \times 0.602 = 15 \text{ m/s}$. - Step 2: Initial vertical component: $v_{0y} = v_0 \sin \theta = 25 \times \sin 53^\circ = 25 \times 0.799 = 20 \text{ m/s}$. - Step 3: Vertical velocity at $t = 2 \text{ s}$: $v_y = v_{0y} - gt = 20 - 9.8 \times 2 = 20 - 19.6 = 0.4 \text{ m/s}$. - Step 4: Since v_y is still positive (moving upward), the ball hasn't reached its peak yet. Confirm: $t_{\text{up}} = v_{0y}/g = 20/9.8 = 2.04 \text{ s}$, which is just after $t = 2 \text{ s}$. **Answer: $v_x = 15 \text{ m/s}$, $v_y = 0.4 \text{ m/s}$ — the ball has not yet reached its highest point**

10. Complementary angles — same range, different time Given: $v_0 = 18 \text{ m/s}$, $\theta_1 = 30^\circ$, $\theta_2 = 60^\circ$. - Step 1 (range at 30°): $R = v_0^2 \sin(2\theta) / g = 18^2 \times \sin 60^\circ / 9.8 = 324 \times 0.866 / 9.8 = 280.6 / 9.8 = 28.6 \text{ m}$. - Step 2 (range at 60°): $R = 324 \times \sin 120^\circ / 9.8$. Since $\sin 120^\circ = \sin 60^\circ = 0.866$, this also gives $R = 28.6 \text{ m}$. - Step 3: Both ranges are equal — confirmed. - Step 4 (time at 30°): $v_{0y} = 18 \sin 30^\circ = 9 \text{ m/s}$, so $T = 2 \times 9 / 9.8 = 1.84 \text{ s}$. - Step 5 (time at 60°): $v_{0y} = 18 \sin 60^\circ = 15.59 \text{ m/s}$, so $T = 2 \times 15.59 / 9.8 = 3.18 \text{ s}$. **Answer: both launches range 28.6 m, but the 60° launch stays airborne almost twice as long (3.18 s vs 1.84 s) — it trades horizontal speed for a higher, longer arc that still lands at the same spot.**